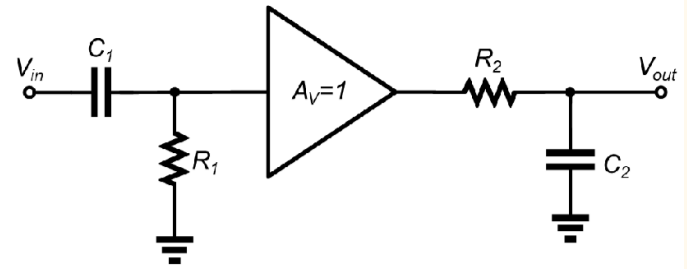


Q1. Fig. 1 shows an amplifier circuit with input and output RC networks.

$C_1=15\text{nF}$, $C_2=560\text{pF}$, $R_1=10\text{k}\Omega$, $R_2=1\text{k}\Omega$

- (1) Derive the circuit transfer function $V_{\text{out}}/V_{\text{in}}$
- (2) Identify the pole and zero in the circuit,
- (3) Sketch the transfer function Bode plot.



Solution:

- Cascade connection of 1st order HPF, amplifier and 1st order LPF

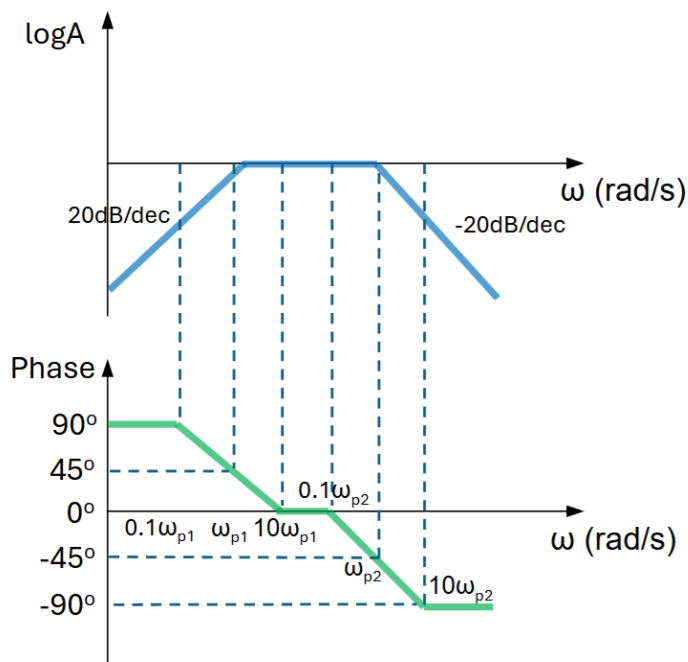
$$(1) H(s) = \frac{s}{s + \frac{1}{R_1 C_1}} \cdot A_V \cdot \frac{\frac{1}{R_2 C_2}}{s + \frac{1}{R_2 C_2}} = \frac{s}{s + \omega_{p1}} \cdot A_V \cdot \frac{\frac{1}{R_2 C_2}}{s + \omega_{p2}} = \frac{s}{s + \omega_{p1}} \cdot \frac{\frac{1}{R_2 C_2}}{s + \omega_{p2}}$$

(2) Zero of the circuit: $\omega_{z1} = 0$

$$\text{Poles of the circuit: } \omega_{p1} = \frac{1}{R_1 C_1} = 6666.7 \text{ rad/s} \quad f_{p1} = \frac{\omega_{p1}}{2\pi} = 1.061 \text{ kHz}$$

$$\omega_{p2} = \frac{1}{R_2 C_2} = 1.78 \times 10^6 \text{ rad/s} \quad f_{p2} = \frac{\omega_{p2}}{2\pi} = 283.4 \text{ kHz}$$

(3) Transfer function Bode Plot



Q2. Fig. 2 shows a two-stage CS amplifier with CSL. $V_{DD}=1.8\text{ V}$, $C_L=10\text{ pF}$, all transistors' $W/L=2\text{ }\mu\text{m}/0.18\text{ }\mu\text{m}$. Parasitic capacitors $C_{gd}=10\text{ fF}$, $C_{gs}=15\text{ fF}$ for all transistors. All transistors work in saturation region. The transistor model parameters are: $V_{THN}=|V_{THP}|=0.4\text{ V}$, $k'_n=200\times 10^{-6}\text{ A/V}^2$, $k'_p=100\times 10^{-6}\text{ A/V}^2$, $\lambda_n=\lambda_p=0.1\text{ V}^{-1}$.

(1) If DC bias current $I_{ref}=10\text{ }\mu\text{A}$ for both stages, calculate the input voltage V_{in} so that V_{out} is 0.9 V . (Consider channel modulation effect)

(2) Continue to calculate the amplifier's DC gain. (You can neglect channel modulation effect)

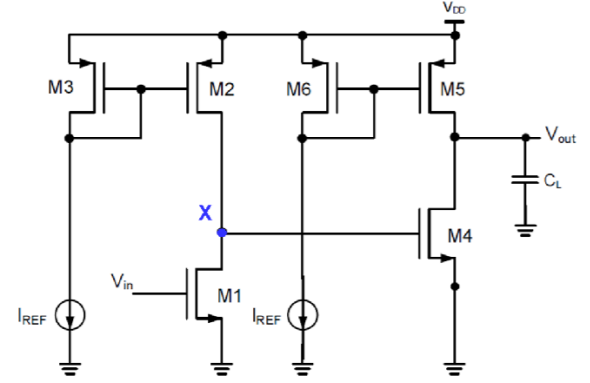
(3) Apply "Association of poles to nodes method" to find frequency of all poles when load $C_L=10\text{ pF}$.

Solution:

For $V_{out} = 0.9\text{ V}$, and all $I_D=10\text{ }\mu\text{A}$,

$$I_{D4} = \frac{k'}{2} \left(\frac{W}{L} \right)_{M4} (V_{GS4} - V_{TH})^2 (1 + \lambda V_{out}) \rightarrow V_{GS4} = 0.4908\text{ V} = V_{DS1}$$

$$I_{D1} = \frac{k'}{2} \left(\frac{W}{L} \right)_{M1} (V_{GS1} - V_{TH})^2 (1 + \lambda V_{DS1}) \rightarrow V_{GS1} = 0.4926\text{ V} = V_{in}$$



- it assumes no gain error

(2) DC Gain

Small signal parameters:

$$g_{m1} = g_{m4} = \sqrt{2k'_n(W/L)I_D} = \sqrt{2k'_n(W/L)I_{REF}} = 210.8 \times 10^{-6}\text{ S}$$

$$r_{o1} = r_{o4} = \frac{1}{\lambda_n I_D} = 1\text{ M}\Omega \quad r_{o2} = r_{o5} = \frac{1}{\lambda_p I_D} = 1\text{ M}\Omega$$

$$\text{Voltage gain of the 1st stage: } r_x = r_{o2} // r_{o1} = 500\text{ k}\Omega \quad A_1 = -g_{m1} r_x = -105.41$$

$$\text{Voltage gain of the 2nd stage: } r_{out} = r_{o4} // r_{o5} = 500\text{ k}\Omega \quad A_2 = -g_{m4} r_{out} = -105.41$$

$$\text{Overall DC gain: } A_{DC} = A_1 A_2 = 11111.268 (\approx 80.91\text{ dB})$$

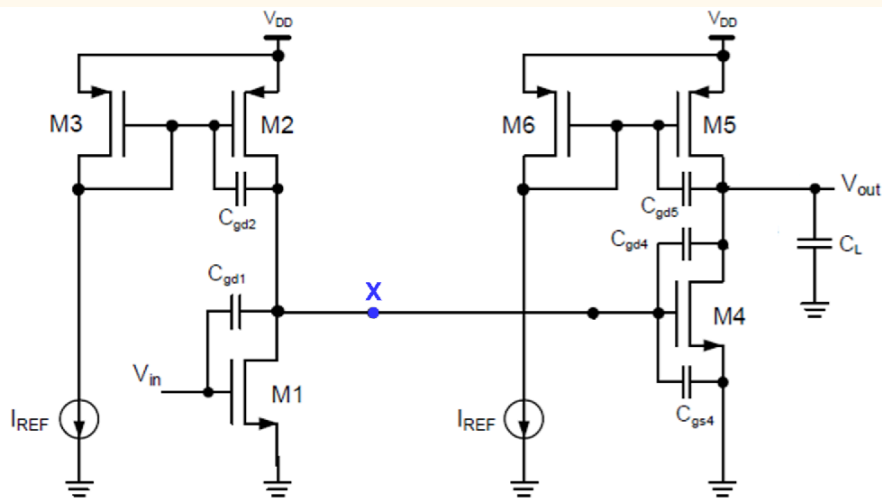
Another Solution:

$$g_{m1} = \frac{2I_D}{V_{GS} - V_{TH}} = 216 \times 10^{-6}\text{ S}$$

$$A_1 = -g_{m1} r_x = -108$$

$$A_2 = -g_{m2} r_{out} = -108$$

- here can see 80 dB gain can be very easily gotten from cascading stages, remember that the subproject 1 is about 40 dB gain



A few notes:

- Identify all parasitic capacitance. Parasitic capacitance is from a node to ground (C_{gd} can be replaced by applying Miller Theorem). Therefore, we can use “Association of poles to nodes” method to find the pole frequencies.
- The Miller capacitance at the gate of M1 needs not to be considered since $R_s = 0$. Therefore, there are only two poles at node x and output node.
- The gates of M2-3 and M5-6 in the two current mirrors are connected to DC voltage since I_{REF} is constant. For AC analysis, the gates can be considered as grounded.

(3) Find all pole frequencies for $C_L = 10$ pF:

At node X:

$$r_x = r_{o2} // r_{o1} = 500 \text{ k}\Omega \quad C_x = C_{gd1} + C_{gd2} + C_{gs4} + (1 + |A_2|) C_{gd4} = 1085 \text{ fF}$$

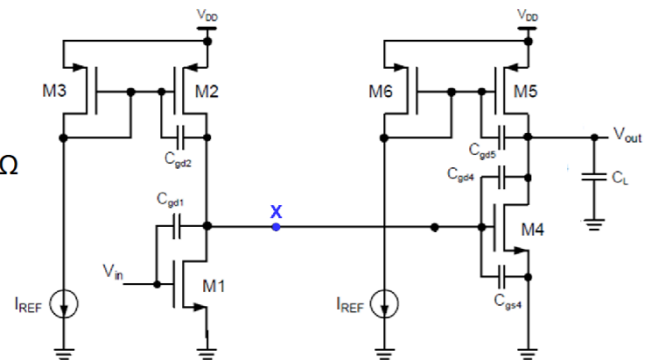
$$\omega_{p1} = \frac{1}{r_x C_x} = 1.852 \times 10^6 \text{ rad/s} \rightarrow f_{p1} = \frac{\omega_{p1}}{2\pi} = 0.295 \text{ MHz}$$

At output node:

$$C_{out} = C_{gd4} + C_{gd5} + C_L = 10 \text{ pF} \quad r_{out} = r_{o4} // r_{o5} = 500 \text{ k}\Omega$$

$$\omega_{p2} = \frac{1}{r_{out} C_{out}} = 0.2 \times 10^6 \text{ rad/s} \rightarrow f_{p2} = \frac{\omega_{p2}}{2\pi} = 31.85 \text{ kHz}$$

Dominant pole



- overall bandwidth is limited by the load

Q3. For a simple current mirror shown below, an NMOS transistor with parameters: $L=1\mu\text{m}$, $V_{TH}=0.6\text{V}$, $k'=200\mu\text{A/V}^2$, $\lambda=0.02\text{V}^{-1}$, $V_{BS}=0\text{V}$, $V_{DD} = 3.0\text{V}$.

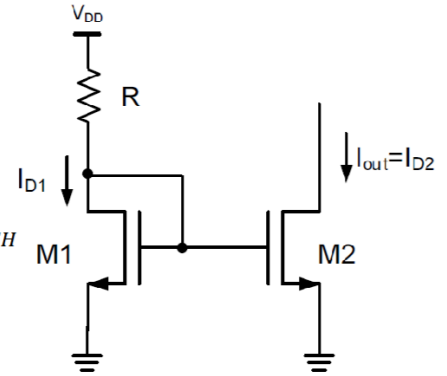
- Given overdrive voltage of 0.3V , calculate R and W of M_1 , so that the drain current of M_1 is $15\mu\text{A}$.
- Design current sources for currents of $7.5\mu\text{A}$, $30\mu\text{A}$, and $45\mu\text{A}$, respectively, and assume that the current mirror is gain error free.
- Find output resistances of current sources in (b) and discuss how the drain current of the current source changes with the drain voltage in $30\mu\text{A}$ case.

Solution:

$$(a) R = \frac{V_{DD} - V_{GS}}{I_{D1}} = \frac{V_{DD} - V_{TH} - V_{OD}}{I_{D1}} = 140\text{k}$$

$$\text{In saturation region: } I_D = \frac{1}{2} k'_n \frac{W}{L} (V_{GS} - V_{TH})^2 (1 + \lambda V_{GS}) \quad V_{OD} = V_{GS} - V_{TH}$$

$$\left(\frac{W}{L}\right)_1 = \frac{2I_{D1}}{k'(V_{OD})^2(1 + \lambda V_{GS})} \approx 1.64 \rightarrow W = 1.64 \mu\text{m}$$



(b) For gain error free, $V_{DS2} = V_{GS}$, $I_{D2} = \frac{(W/L)_2}{(W/L)_1} I_{D1}$

$$I_{D2} = 7.5 \mu\text{A}: \quad \left(\frac{W}{L}\right)_2 = \frac{7.5}{15} \left(\frac{W}{L}\right)_1 \approx 0.82 \rightarrow W = 0.82 \mu\text{m}$$

$$I_{D2} = 30 \mu\text{A}: \quad \left(\frac{W}{L}\right)_2 = \frac{30}{15} \left(\frac{W}{L}\right)_1 \approx 3.28 \rightarrow W = 3.28 \mu\text{m}$$

$$I_{D2} = 45 \mu\text{A}: \quad \left(\frac{W}{L}\right)_2 = \frac{45}{15} \left(\frac{W}{L}\right)_1 \approx 4.92 \rightarrow W = 4.92 \mu\text{m}$$

$$(c) r_o = \frac{1}{\lambda I_{D2}} = \begin{cases} 6.67\text{M}\Omega & (I_{D2} = 7.5 \mu\text{A}) \\ 1.67\text{M}\Omega & (I_{D2} = 30 \mu\text{A}) \\ 1.11\text{M}\Omega & (I_{D2} = 45 \mu\text{A}) \end{cases}$$

$$\text{Change of } I_D \text{ with } V_{DS} \quad I_{D1} = \frac{k'}{2} \left(\frac{W}{L}\right)_1 (V_{OD})^2 (1 + \lambda V_{DS2})$$

$$\frac{\partial I_{D1}}{\partial V_{DS2}} = \lambda \frac{k'}{2} \left(\frac{W}{L}\right)_1 (V_{OD})^2 \approx 0.59 \mu\text{A/V}$$

The output current will increase $0.59 \mu\text{A}$ or $\sim 2\%$ for 1V increase in output voltage